IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

EXAMINATIONS 2025

BEng Honours Degree in Computing Part I
MEng Honours Degrees in Computing Part I
for Internal Students of the Imperial College of Science, Technology and Medicine

This paper is also taken for the relevant examinations for the Associateship of the City and Guilds of London Institute

PAPER COMP40018A

DISCRETE MATHEMATICS

Friday 16th May 2025, 14:00
Duration: 40 minutes

Answer ALL ONE questions

THIS IS A CLOSED BOOK EXAM

Paper contains 1 question
Calculators not required

1 a Let $V = \{a, b, \{b\}\}$ and $W = \{a, \{a\}, c\}$. Determine the *six* sets $V \cap W$, $V \cup W$, $V \setminus W$, $\wp W$, $V \cap \wp W$, and $V \bigtriangleup W$.

b
i) Is a reflexive relation always symmetric?
ii) Is a symmetric relation always reflexive?
iii) Is the union of two symmetric relations always symmetric?
iv) Is the complement of a transitive relation a transitive relation?

Give a proof when the statement is true, and a counterexample with explanation if it is not.

c
i) Let R be a partial order on A. Give the definitions of *minimal*, *least*. *maximal*, and *greatest* element in A with respect to R.

ii) We define a collection of subsets of $\mathbb{N}$, a subset W of $\wp\mathbb{N}$, and a binary relation R on $\mathbb{N}$ through:

$$\begin{array}{rcl} V_k & \triangleq & \{n \in \mathbb{N} \mid \exists\, m \in \mathbb{N}\,(m > 0 \wedge k \times m = n)\} \\ W & \triangleq & \{V_k \in \wp\mathbb{N} \mid k \in \mathbb{N}\} \\ q\,R\,p & \triangleq & V_p \subseteq V_q \end{array}$$

Does the relation R have minimal, least, maximal, or greatest elements? If yes, are they unique? If no, argue why. Give examples, and show why each satisfies the criterion.

d Determine which of the following relations are (partial) functions. For those which are functions,

- determine whether they are injective and/or surjective;
- give the inverse function when it exists.

Justify your answers throughout.

i) $\{\langle x, y\rangle \in \mathbb{Z}^2 \mid x = y \wedge x = -y\}$;
ii) $\{\langle x, y\rangle \in (\mathbb{R} \setminus \{2\}) \times (\mathbb{R} \setminus \{0\}) \mid y = 1/(x^3 - 8)\}$;
iii) $\{\langle x, y\rangle \in \mathbb{R}^2 \mid x^2 = y^3\}$.
iv) $\{\langle x, y\rangle \in \mathbb{Q}^2 \mid y = x^3\}$.

e Prove that if V is not countable, and $V \subseteq W$, then neither is W. State clearly in detail which results from the notes you use in your proof.

The five parts carry, respectively, 15%, 20%, 30%, 20%, and 15% of the marks.